## Corrected MCQs

1. **B.** The fundamental theorem gives $F(b) - F(a)$. It applies to a antiderivative of $f$, not to $f$ She herself.

2. **C.** For $a < b$, the integral retains the sign of a negative or null function.
   It's either positive or null.
3. **A.** The integral is divided by the length of the interval $b - a > 0$. For a constant function,
   Retrieve this constant.
4. **B.** Areas on constant sign intervals are calculated and their positive values are added.
   Without cutting, areas can be compensated.
5. **B.** The derivative of $e^x$ is $e^x$. The other functions proposed have different derivatives.
6. **A.** The usual formula of IPP applies to two class functions $\mathcal{C}^1$ on the interval.
   guarantees the continuous derivatives and integrals used.
7. **A.** The formula comes from $(uv)' = u'v + uv'$. Integrate and isolate the first term gives $\int_a^b u'v =$
   $[uv]_a^b - \int_a^b uv'$.
8. **B.** A increasing function is above the height taken at the left end of each
   The rectangles on the left provide a minor.
9. **A.** For $x \in [0;1]$, $0 \leq x^n \leq 1$ ; one can multiply by $f(x) \geq 0$ then integrate. The bound can be
   refined by a upper bound of $f$.
10. **B.** The product of a flow in litres per minute by a duration in minutes is expressed in litres.
    The integral represents a cumulative volume.

**True/false.** False: $\int_0^1 (-1)\,\mathrm{d}x = -1$. True: the difference of antiderivatives has a zero derivative on a interval, therefore, is constant. False: growth or decay determines which side gives the Minorizing.

## Corrected applications and support

**Application 1.** $g = 2f$ : integral 3, area 5. The factor 2 is positive, so the signs are preserved and both areas are doubled.

**Application 2.** $A(x) = 1 - e^{-x}$, $A' = e^{-x} > 0$, therefore growth and limit 1. $e^{-t}$ between 0 and x increases and approaches 1.

**Implementation 3.** $m = 1/3$. The rectangle has width 1 and height 1/3. The average of the ends would be 1/2: it measures the top trapeze area, not the one under the curve.

**Implementation 4.** $L_4 = \frac{1}{4}(0 + 1/16 + 1/4 + 9/16) = 7/32$ ; $U_4 = 15/32$. Growth $L_4 \leq 1/3 \leq U_4$. $U_n - L_n = \frac{1}{n}[1^2 - 0^2] = 1/n$.

**Application 5.** Take u=x and $v = \ln x$, therefore $u' = 1$, $v' = 1/x$. The integral value is $[x \ln x]_1^e - \int_1^e 1 \, \mathrm{d}x = e - (e - 1) = 1$. A antiderivative is $x \ln x - x$, whose derivative is ln $x$.

**Implementation 6.** On [0;1], 0 $\leq x^{n+1} \leq x^n$ ; multiply by $e^{-x} > 0$ then integrate proves positivity and decrease. From $e^{-1} \leq e^{-x} \leq 1$, we get the bounds, then $I_n \to 0$. A PPI applied to $I_{n+1}$ gives $I_{n+1} = -e^{-1} + (n+1)I_n$. This relationship is accurate but is not enough alone to calculate the limit.

**Implementation 7.** $f(x) = x$ If f is continuous and positive or zero, a value strictly positive would result by continuity a strictly positive area over a small interval, contradiction. Thus a positive continuous function of zero integral is identically null.

**Splitting of the meeting**

Separate the installation from the IPP signed area. Work on rectangles is done first on the then in sum. In the integral sequence, let choose between monotonous convergence and framework; the second provides the value of the limit here directly.

## Conduct and correction

**Benchmarks for the teacher**

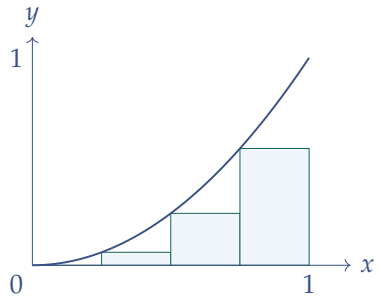
Start with a positive area, then show an example including positive and negative areas
For an average value, require a unit and an interpretation.
Integrals arrive in S66; reserve a real search phase at the choice of comparison.
The evidence of the fundamental theorem is limited to the increasing positive case indicated in the program.
Trigonometric calculations will be repeated after May.

### Corrected activity

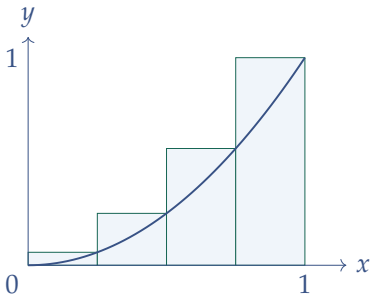
1. The four intervals are wide $\frac{1}{4}$. The sum on the left is

$$L_4 = \frac{1}{4}(0^2 + (\frac{1}{4})^2 + (\frac{1}{2})^2 + (\frac{3}{4})^2) = \frac{7}{32}.$$

Right, $U_4 = \frac{1}{4}((\frac{1}{4})^2 + (\frac{1}{2})^2 + (\frac{3}{4})^2 + 1) = \frac{15}{32}$. The rectangles have for heights the values at corresponding ends.




Left rectangles: $L_4 = \frac{7}{32}$.




Right rectangles: $U_4 = \frac{15}{32}$.

2. For $x \in [\frac{k}{4}, \frac{k+1}{4}]$, $(\frac{k}{4})^2 \le x^2 \le (\frac{k+1}{4})^2$ because the square function is increasing on $[0,1]$. Integration on the four intervals then the addition give $\frac{7}{32} \le I \le \frac{15}{32}$.

3.

```
def rectangles_square(n):
    h = 1/n
    Low = h* sum((k*h)**2 for k in rows (n))
    high = h* sum((k*h)**2 for k in rows (1, n+1))
    return low, high
```

Domestic terms compensate for: $U_n - L_n = (\frac{1}{n})(1^2 - 0^2) = \frac{1}{n}$.

4. $\frac{1}{n} \le 0{,}01 \iff n \ge 100$. One hundred subdivisions are sufficient; the first integer gives this width with these two sums.

5. A antiderivative of $x^2$ is $\frac{x^3}{3}$. So $I = [\frac{x^3}{3}]_0^1 = \frac{1}{3}$, included between $\frac{7}{32}$ and $\frac{15}{32}$.

**E1.** A antiderivative is $F(x) = x^3 - x^2 + x$. The integral value is $F(2) - F(0) = 8 - 4 + 2 = 6$.

**E2.** On $[0,1]$, $x - x^2 = x(1-x) \ge 0$, so the right is above the parabola. $\int_0^1 (x - x^2)\,dx = [\frac{x^2}{2} - \frac{x^3}{3}]_0^1 = \frac{1}{6}$ Dairy unit.

**E3.** The length of the interval is 2. $(\frac{1}{2})\int_0^2 e^x\,dx = (\frac{1}{2})[e^x]_0^2 = \frac{e^2-1}{2}$.

**E4.** Choices $u = \ln x$, $u' = \frac{1}{x}$, $v' = 1$ and $v = x$ on $[1, e]$. IPP gives $\int_1^e \ln x \, dx = [x \ln x]_1^e - \int_1^e 1 \, dx = e - (e-1) = 1$.

**E5.** On $[0, 1]$, $1 + x \geq 1$ and $x^n \geq 0$, therefore $0 \leq \frac{x^n}{1+x} \leq x^n$. Integration gives $0 \leq I_n \leq [\frac{x^{n+1}}{n+1}]_0^1 = \frac{1}{n+1}$. The gendarmes give $I_n \to 0$.

**Corrected from DM8**

1. $I_0 = \int_0^1 e^x \, dx = e - 1$. For $I_1$a PPI with $u = x$, $v' = e^x$ gives $I_1 = [xe^x]_0^1 - \int_0^1 e^x \, dx = e - (e-1) = 1$.

2. For $n \geq 0$, choose $u = x^{n+1}$, $u' = (n+1)x^n$, $v' = e^x$ and $v = e^x$. The functions are derivatable With continuous derivatives on $[0, 1]$. Then

$$I_{n+1} = [x^{n+1}e^x]_0^1 - (n+1)\int_0^1 x^n e^x \, dx = e - (n+1)I_n.$$

3. On $[0, 1]$, $0 \leq x^{n+1} \leq x^n$. By multiplying by $e^x > 0$ then integrating, $0 \leq I_{n+1} \leq I_n$ : the rest is positive and decreasing in the broad sense. It is even strictly positive and strictly decreasing, for the integrands of $I_n$ and $I_n - I_{n+1}$ are strictly positive on $]0, 1[$.

4. Like $1 \leq e^x \leq e$ and $x^n \geq 0$ on $[0, 1]$, we can integrate $x^n \leq x^n e^x \leq ex^n$. Where

$$\frac{1}{n+1} \leq I_n \leq \frac{e}{n+1}.$$

The two bounds tend towards 0; the theorem of the gendarmes gives $I_n \to 0$.

5. If $\widetilde{I}_n = I_n + \varepsilon_n$ and that the relationship is then applied without further error, then

$$\widetilde{I}_{n+1} - I_{n+1} = -(n+1)(\widetilde{I}_n - I_n) = -(n+1)\varepsilon_n.$$

The absolute value of the error is multiplied by $n + 1$. In machine calculation, new errors The decrease in the exact integrals therefore does not guarantee stability. of this method of calculation.

**Corrected work in the manual or an anal**

**Fiscal year 28.** On $[0; 1]$, let's choose $u(x) = x$ and $v(x) = e^x$. These functions are class $\mathcal{C}^1$, $u'(x) = 1$ and $v'(x) = e^x$. Part-by-part integration gives

$$\int_0^1 xe^x \, dx = [xe^x]_0^1 - \int_0^1 e^x \, dx = e - (e-1) = 1.$$

**Fiscal year 29.** On $[1; e]$, let's choose $u(x) = \ln x$ and $v(x) = \frac{x^2}{2}$. Then $u'(x) = \frac{1}{x}$ and $v'(x) = x$ ; these posts are at the grade level $\mathcal{C}^1$. Where

$$\int_1^e x \ln x \, dx = \left[\frac{x^2}{2}\ln x\right]_1^e - \frac{1}{2}\int_1^e x \, dx$$
$$= \frac{e^2}{2} - \frac{e^2-1}{4} = \frac{e^2+1}{4}.$$